

Common-Digit Fabius Zonoids

Exact volumes, hyperbolic-secant geometry,
Bernoulli Gaussianization, and parameter jets

A repository-audited frontier report for `ProveIt/Analysis/FabiusFunction/docs`

The central construction couples every geometric-uniform/Fabius parameter through the *same* independent uniform digits. This produces an infinite-dimensional random analytic curve whose finite parameter samples have compact zonoid supports. The report derives exact support volumes, mixed Bernoulli cumulants, a hidden Szegő/Hardy-space geometry, a hyperbolic-secant covariance kernel, Gaussian limits, Euler-number determinant identities, and confluent jet volumes.

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Research status: rigorous derivations plus explicitly labeled conjectures; numerical experiments are checks only.

Abstract

Let $(U_n)_{n \geq 0}$ be independent uniform random variables on $[0, 1]$ and, for $0 < q < 1$, put

$$X_q = (1 - q) \sum_{n \geq 0} q^n U_n.$$

At $q = \frac{1}{2}$ this is the standard probabilistic realization of the Fabius/Rvachev law used throughout the source repository. Existing work studies each q separately and develops its q -series, sinc-product, asymptotic, inverse-function, Legendre, Bell, Bernoulli, and Thue–Morse structures. Here the same digit sequence is used for several parameters simultaneously.

For distinct $0 < q_1 < \dots < q_d < 1$, the support of $(X_{q_1}, \dots, X_{q_d})$ is the infinite zonotope generated by

$$v_n = ((1 - q_1)q_1^n, \dots, (1 - q_d)q_d^n).$$

Its exact volume is

$$\text{Vol}_d Z(q_1, \dots, q_d) = \prod_{i < j} \frac{q_j - q_i}{1 - q_i q_j}.$$

Writing $q_i = \tanh u_i$ turns this into $\prod_{i < j} \tanh(u_j - u_i)$. After standardization, the covariance kernel becomes $\text{sech}(u - v)$, and a Cauchy determinant gives the unexpected identity

$$\text{Vol}_d Z = \sqrt{\det[\text{Corr}(X_{q_i}, X_{q_j})]}.$$

The same kernel is the normalized Szegő kernel on the disk. It yields exact best-linear-prediction errors, total-positivity determinants, and a confluent Euler-number identity

$$\det[\mu_{i+j}]_{i,j=0}^{d-1} = \prod_{k=0}^{d-1} (k!)^2,$$

where $\mu_{2m} = |E_{2m}|$ and $\mu_{2m+1} = 0$ are the moments of the hyperbolic-secant law.

All joint cumulants are explicit Bernoulli rational functions. Hyperbolic translation $u \mapsto T + u$ suppresses every cumulant of order $m > 2$ at rate $e^{-(m-2)T}$, proving convergence to the stationary Gaussian process with covariance $\text{sech}(u - v)$ and providing a Bell-partition expansion beyond Wick's rule. Divided-difference confluence gives the exact support volume of the parameter jet

$$(X_q, X'_q, X''_q/2!, \dots, X_q^{(d-1)})/(d-1)!$$

as $(1 - q^2)^{-\binom{d}{2}}$. Further sections give the exact two-parameter geometric-comb boundary, inverse-Fabius copula formulas, shifted-Legendre coefficients, a Thue–Morse vector filter, Lambert- W truncation scales, numerical checks, conjectures, and a formalization roadmap.

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1 Repository audit, scope, and novelty boundary

1.1 How the source corpus was used

The requested source tree is large, layered, and intentionally redundant [1, 2]. It contains a polished primary synthesis, a canonical semi-formalized frontier compilation, topic-specific papers, a manifest of drafts, historical archives, and links or copies retained for provenance. Reading every duplicate as though it were independent would obscure rather than improve a novelty check. The audit therefore used the following hierarchy.

- A1. The live primary synthesis was treated as the authoritative baseline for definitions, normalizations, proved one-parameter identities, and the Fabius/Rvachev correspondence.
- A2. The canonical semi-formalized frontier document and its thematic manifest were used to map claimed extensions: exact dyadic values, Thue–Morse transforms, q -Pochhammer and q -binomial forms, infinite sinc products, endpoint Lambert- W asymptotics, inverse Fabius expansions, Legendre/Bell/Bernoulli machinery, interpolation, total positivity, fractional calculus, and related directions.
- A3. Topic-specific drafts were checked where their titles or manifest descriptions intersected the present construction, especially the recent parameter-susceptibility report.
- A4. Archived duplicates were used as provenance checks, not counted as new mathematical content when their own archive notes identified them as superseded or consolidated.
- A5. Finally, targeted full-text searches were made for the decisive notions introduced below: common-digit or common-noise coupling across several q 's, joint characteristic functions, mixed cumulants, zonotopes/zonoids, support volume, Cauchy/Szegő correlation determinants, hyperbolic-secant covariance, and confluent multivariate support geometry.

This is a *repository novelty* protocol. It supports the statement that the central results below were not found in the active corpus or its manifest under equivalent formulations. It does not assert absolute historical priority over all of mathematics. Some ingredients—zonotope volume formulas, Schur identities, Cauchy determinants, Prékopa closure, and Meixner–Pollaczek orthogonality—are classical and are cited explicitly.

Existing repository input

The report imports the following one-parameter facts rather than re-proving the repository's full theory: the geometric-uniform realization X_q ; the identification of $q = 1/2$ with the normalized Fabius/Rvachev law; the marginal infinite sinc product; the role of Bernoulli cumulants and Bell polynomials; exact dyadic/Thue–Morse identities; and the established endpoint and inverse-Fabius asymptotic framework. These are background inputs, not claimed contributions of this report.

1.2 Main contributions and status

Theme	Result	Status
Common-digit lift	A multivariate Fabius family with an explicit joint sinc product and all mixed Bernoulli cumulants.	proved here
Support geometry	Compact full-dimensional infinite zonotope; C_c^∞ log-concave associated density.	proved here

Theme	Result	Status
Exact volume	$\text{Vol } Z = \prod_{i < j} (q_j - q_i) / (1 - q_i q_j)$.	proved here
Hyperbolic geometry	$q = \tanh u$ converts volume factors to tanh distances and correlations to sech distances.	proved here
Support–correlation identity	$\text{Vol } Z = \sqrt{\det R}$, with exact linear-prediction errors and Szegő-kernel interpretation.	proved here
Two-parameter lens	Exact geometric-comb boundary and log-periodic interpolation of $x^{\log r / \log q}$.	proved here
Gaussianization	Hyperbolic translations converge to the stationary Gaussian sech process; all cumulant corrections are explicit.	proved here
Euler jets	Euler Hankel determinant, Meixner–Pollaczek recurrence, and a unit-variance local derivative-innovation ladder.	proved here
Confluent support	Exact parameter-jet and multi-jet support volumes, plus Lambert- W cutoff scales.	proved here
Inverse/Legendre bridges	Exact copula support using inverse Fabius; rational mixed shifted-Legendre coefficients.	proved here
Signed/complex/ray asymptotics	Chamber formulas, complex Pick geometry, sharp joint sinc-product decay, and endpoint copula laws.	conjecture

2 The one-parameter geometric-uniform law

2.1 Definition and elementary structure

Let $U_0, U_1, \dots$ be independent random variables, each uniformly distributed on $[0, 1]$. For $q \in (0, 1)$ define

$$X_q = (1 - q) \sum_{n=0}^{\infty} q^n U_n. \quad (1)$$

The coefficients are positive and sum to one, so $0 \leq X_q \leq 1$. The series converges absolutely for every digit sequence, uniformly on compact subsets of $|q| < 1$, and therefore $q \mapsto X_q$ extends samplewise to a real-analytic function on $(-1, 1)$.

The reflection $U_n \mapsto 1 - U_n$ gives

$$X_q \stackrel{d}{=} 1 - X_q. \quad (2)$$

In particular $\mathbb{E}X_q = 1/2$. Direct summation gives

$$\text{Var}(X_q) = \frac{1}{12} (1 - q)^2 \sum_{n \geq 0} q^{2n} = \frac{1 - q}{12(1 + q)}. \quad (3)$$

For $z \in \mathbb{C}$, write $\text{sinc}_{\text{rad}} z = \sin z / z$ with $\text{sinc}_{\text{rad}} 0 = 1$. The characteristic function is

$$\phi_q(t) = \exp(it/2) \prod_{n \geq 0} \text{sinc}_{\text{rad}} \left(\frac{t(1 - q)q^n}{2} \right). \quad (4)$$

This is the familiar infinite sinc product. The repository develops its special $q = 1/2$ behavior and its relation to the Fourier transform of Rvachev’s up-function in detail.

2.2 Fabius normalization at $q = 1/2$

Let $F_q(x) = \mathbb{P}(X_q \leq x)$ and let f_q denote its density. Under the normalization used in the repository,

$$F_{1/2} = F, \quad (5)$$

where F is the Fabius function on $[0, 1]$ [3, 5, 6]. Equivalently, if $Y = 2X_{1/2} - 1$, then Y has Rvachev up-density on $[-1, 1]$, and

$$f_{1/2}(x) = 2 \operatorname{up}(2x - 1). \quad (6)$$

This convention will matter in [section 10](#): the quantile $F_{1/2}^{-1}$ is precisely the inverse Fabius function.

2.3 Bernoulli cumulants

Let B_m^+ denote the Bernoulli convention with $B_1^+ = 1/2$ and $B_m^+ = B_m$ for $m \neq 1$. Since

$$\log \mathbb{E} e^{tU_0} = \log \frac{e^t - 1}{t} = \sum_{m \geq 1} \frac{B_m^+}{m} \frac{t^m}{m!},$$

the m th cumulant of U_0 is B_m^+/m . Additivity and homogeneity of cumulants yield

$$\kappa_m(X_q) = \frac{B_m^+}{m} \frac{(1-q)^m}{1-q^m}. \quad (7)$$

For $m > 1$ odd this vanishes. Formula (7), followed by the complete Bell-polynomial moment-cumulant relation, is the one-parameter precursor of the multivariate calculus below.

3 The common-digit multivariate lift

3.1 Definition and support

Fix distinct parameters

$$0 < q_1 < q_2 < \cdots < q_d < 1 \quad (8)$$

and use the *same* uniforms in all coordinates. Put

$$\mathbf{X}_Q = (X_{q_1}, \dots, X_{q_d}), \quad v_n = ((1 - q_1)q_1^n, \dots, (1 - q_d)q_d^n)^\top. \quad (9)$$

Then

$$\mathbf{X}_Q = \sum_{n \geq 0} U_n v_n. \quad (10)$$

The deterministic support candidate is the convergent Minkowski sum

$$Z_Q = \sum_{n \geq 0} [0, v_n] = \left\{ \sum_{n \geq 0} u_n v_n : 0 \leq u_n \leq 1 \right\}. \quad (11)$$

Here convergence is in Hausdorff distance because $\sum_n \|v_n\| < \infty$.

Theorem 3.1 (Common-digit support zonoid). *For (8), $\operatorname{supp}(\mathbf{X}_Q) = Z_Q$. The set Z_Q is a compact, convex, full-dimensional zonoid contained in $[0, 1]^d$, centrally symmetric about $(1/2, \dots, 1/2)$.*

Proof. The product measure on $[0, 1]^{\mathbb{N}_0}$ has full support. The map $\mathbf{u} \mapsto \sum_n u_n v_n$ is continuous for the product topology restricted through the summable coefficient bound, so the support of its pushforward is the image of the product support, namely (11). Compactness and convexity follow from Hausdorff limits of finite zonotopes. Since $\sum_n v_n = (1, \dots, 1)$, replacing every u_n by $1 - u_n$ reflects the set through its stated center.

It remains to prove nonempty interior. The first d generators form a matrix with determinant

$$\det[v_0 \cdots v_{d-1}] = \left(\prod_{i=1}^d (1 - q_i) \right) \prod_{1 \leq i < j \leq d} (q_j - q_i) > 0. \quad (12)$$

Thus the finite partial sum generated by $v_0, \dots, v_{d-1}$ already contains a full-dimensional parallelepiped. $\square$

3.2 Joint infinite sinc product and its zero hyperplanes

Theorem 3.2 (Joint characteristic function). *For $t = (t_1, \dots, t_d) \in \mathbb{R}^d$,*

$$\Phi_Q(t) = \exp\left(\frac{i}{2} \sum_{j=1}^d t_j\right) \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{1}{2} \sum_{j=1}^d t_j (1 - q_j) q_j^n\right). \quad (13)$$

Its real zero set is exactly the locally finite union of affine-through-the-origin hyperplanes

$$\sum_{j=1}^d t_j (1 - q_j) q_j^n = 2\pi k, \quad n \geq 0, \quad k \in \mathbb{Z} \setminus \{0\}. \quad (14)$$

Proof. Independence gives one uniform characteristic factor for each n . Their phase factors multiply to $\exp(it \cdot \sum_n v_n / 2) = \exp(i \sum_j t_j / 2)$, proving (13). For fixed t , the arguments in the tail are exponentially small and their squares are summable. Since $\log \text{sinc}_{\text{rad}} z = -z^2/6 + O(z^4)$ near zero, the tail product converges to a nonzero number. Therefore the product vanishes precisely when one finite factor vanishes, giving (14). $\square$

Remark 3.3. Along a ray $t = s\theta$, the factor arguments are values of an exponential polynomial

$$A_n(\theta) = \sum_j \theta_j (1 - q_j) q_j^n.$$

This turns multivariate Fourier decay into a problem about dominance and cancellation among several geometric scales. It is the natural common-digit extension of the one-parameter infinite-sinc decay problem; a sharp conjectural stratification is stated in theorem 14.3.

3.3 All mixed cumulants

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d \setminus \{0\}$, set $|\alpha| = \sum_j \alpha_j$. Let $\kappa_\alpha(\mathbf{X}_Q)$ denote the joint cumulant in which coordinate j occurs α_j times.

Theorem 3.4 (Mixed Bernoulli cumulant formula). *For $m = |\alpha|$,*

$$\kappa_\alpha(\mathbf{X}_Q) = \frac{B_m^+}{m} \frac{\prod_{j=1}^d (1 - q_j)^{\alpha_j}}{1 - \prod_{j=1}^d q_j^{\alpha_j}}. \quad (15)$$

After centering, every joint cumulant of odd total order vanishes.

Proof. The contribution of digit n is the scalar U_n multiplied by each required coefficient. Its mixed cumulant is

$$\kappa_m(U_n) \prod_j ((1 - q_j) q_j^n)^{\alpha_j}.$$

Joint cumulants add over independent n , and the remaining sum is geometric. Symmetry of $U_n - 1/2$ eliminates odd orders. $\square$

Formula (15) is already a complete symbolic engine. If $i_1, \dots, i_m \in \{1, \dots, d\}$ and $\mathcal{P}([m])$ denotes the set of set partitions, then

$$\mathbb{E} \prod_{a=1}^m X_{q_{i_a}} = \sum_{\pi \in \mathcal{P}([m])} \prod_{B \in \pi} \kappa(X_{q_{i_a}} : a \in B). \quad (16)$$

Thus every mixed moment is a finite Bell-partition expression in rational functions of the q_j . In particular, rational parameters give rational moments exactly.

3.4 Smooth density, flat boundary, log-concavity, and association

Theorem 3.5 (Regularity and shape). *The law of $\mathbf{X}_Q$ has a density $f_Q \in C_c^\infty(\mathbb{R}^d)$ supported on Z_Q . Every derivative of f_Q vanishes on ∂Z_Q . Moreover,*

$$\|f_Q\|_\infty \leq \frac{1}{\prod_{i=1}^d (1 - q_i) \prod_{i < j} (q_j - q_i)}. \quad (17)$$

The measure is log-concave and associated: for bounded coordinatewise nondecreasing A, B ,

$$\text{Cov}(A(\mathbf{X}_Q), B(\mathbf{X}_Q)) \geq 0. \quad (18)$$

Proof. The sum of the first d digit vectors has the uniform density on the parallelepiped generated by $v_0, \dots, v_{d-1}$, with essential supremum equal to the reciprocal of (12). Convolution with the remaining probability measure cannot increase the L^∞ norm, proving (17).

For smoothness, partition arbitrarily many generators into disjoint blocks of d consecutive vectors. Every block matrix is nonsingular, because after factoring positive row and column powers it is again a Vandermonde matrix. For one such block, at least one scalar product $t \cdot v_n$ is bounded below by a block-dependent positive constant times $\|t\|$; hence its characteristic product is $O(\|t\|^{-1})$. Using M disjoint blocks gives $\Phi_Q(t) = O_M(\|t\|^{-M})$ for every M . Fourier inversion then gives $f_Q \in C_c^\infty$. Since every derivative is continuous on all of $\mathbb{R}^d$ and is zero in the exterior of Z_Q , it is zero on the boundary as well.

Each finite digit projection is the linear image of a uniform measure on a cube, hence log-concave by Prékopa's theorem [12]; log-concavity is preserved under weak limits. Finally every coordinate X_{q_j} is an increasing function of every independent U_n . The Harris association inequality for product measures gives (18) for finite truncations, and bounded convergence passes it to the limit. $\square$

4 Exact zonoid volume

The support volume is the first genuinely multivariate invariant of the construction. Its closed form is unexpectedly simple.

4.1 Finite zonotopes and generalized Vandermonde minors

Let

$$Z_Q^{(N)} = \sum_{n=0}^N [0, v_n].$$

The standard zonotope formula [8, 9] gives

$$\text{Vol}_d Z_Q^{(N)} = \sum_{0 \leq n_1 < \dots < n_d \leq N} |\det[v_{n_1} \cdots v_{n_d}]|. \quad (19)$$

For ordered positive q_i , every displayed determinant has the same sign. Factoring the rows gives

$$\det[v_{n_1} \cdots v_{n_d}] = \left(\prod_{i=1}^d (1 - q_i) \right) \det[q_i^{n_j}]_{i,j=1}^d. \quad (20)$$

Given $0 \leq n_1 < \dots < n_d$, define the partition

$$\lambda_j = n_{d-j+1} - (d - j), \quad 1 \leq j \leq d. \quad (21)$$

The generalized Vandermonde identity is

$$\det[q_i^{n_j}] = \prod_{i < j} (q_j - q_i) s_\lambda(q_1, \dots, q_d), \quad (22)$$

where s_λ is a Schur polynomial. The relevant Littlewood identity [10, 11] is

$$\sum_{\lambda} s_\lambda(q_1, \dots, q_d) = \frac{1}{\prod_{i=1}^d (1 - q_i) \prod_{i < j} (1 - q_i q_j)}. \quad (23)$$

The sum ranges over all partitions of length at most d and converges absolutely for $|q_i| < 1$.

4.2 The volume product

Theorem 4.1 (Exact common-digit support volume). *For $0 < q_1 < \dots < q_d < 1$,*

$$\boxed{\text{Vol}_d Z_Q = \prod_{1 \leq i < j \leq d} \frac{q_j - q_i}{1 - q_i q_j}}. \quad (24)$$

Proof. As $N \rightarrow \infty$, the finite zonotopes increase to Z_Q in Hausdorff distance and their volumes converge to $\text{Vol } Z_Q$. In (19), use (20) and (22); then let $N \rightarrow \infty$. The index map (21) exhausts all partitions of length at most d . Formula (23) cancels the factor $\prod_i (1 - q_i)$ and leaves exactly (24). $\square$

Proved in this report

The decisive mechanism is a three-step bridge:

$$\text{uniform digits} \rightarrow \text{zonotope minors} \rightarrow \text{Schur-Littlewood sum}.$$

Neither the probabilistic definition alone nor the marginal q -series reveals the pairwise pseudohyperbolic product in (24).

Corollary 4.2 (Basic bounds and collision order). *One has $0 < \text{Vol } Z_Q < 1$. If $q_j = q + \varepsilon s_j$ with distinct fixed $s_1 < \dots < s_d$, then, as $\varepsilon \rightarrow 0^+$,*

$$\text{Vol } Z_Q = \frac{\varepsilon^{\binom{d}{2}} \prod_{i < j} (s_j - s_i)}{(1 - q^2)^{\binom{d}{2}}} (1 + O(\varepsilon)). \quad (25)$$

Proof. Every factor in (24) lies in $(0, 1)$. The collision formula follows by expanding each numerator and denominator. $\square$

4.3 Equally spaced hyperbolic parameters and a finite q -superfactorial

The product has a particularly transparent q -analog form for a hyperbolic arithmetic progression. Let

$$q_j = \tanh(u_0 + (j - 1)h), \quad h > 0,$$

and put $Q = e^{-2h}$. Then [theorem 4.1](#) and the hyperbolic subtraction law give

$$\text{Vol } Z_Q = \prod_{k=1}^{d-1} \tanh(kh)^{d-k} = \prod_{j=1}^{d-1} \frac{(Q; Q)_j}{(-Q; Q)_j}, \quad (26)$$

where $(a; Q)_j = \prod_{k=0}^{j-1} (1 - aQ^k)$. The volume is independent of the base point u_0 .

Define

$$\rho(h) = \prod_{k \geq 1} \tanh(kh) = \frac{(Q; Q)_\infty}{(-Q; Q)_\infty}, \quad C(h) = \prod_{k \geq 1} \tanh(kh)^{-k}. \quad (27)$$

Absolute convergence of the logarithms gives the large- d law

$$\text{Vol } Z_Q = C(h) \rho(h)^d \left(1 + O(dQ^d)\right). \quad (28)$$

Using $(-Q; Q)_\infty = (Q^2; Q^2)_\infty / (Q; Q)_\infty$ and the modular asymptotic of the Dedekind eta function yields, as $h \downarrow 0$,

$$\rho(h) = \sqrt{\frac{2\pi}{h}} \exp\left(-\frac{\pi^2}{8h}\right) \left(1 + O(e^{-\pi^2/h})\right). \quad (29)$$

Thus dense hyperbolic sampling drives high-dimensional support volume down at a modularly controlled exponential rate. The still subtler small- h expansion of $C(h)$ leads naturally to q -Barnes and MacMahon-type products; see [theorem 14.7](#).

5 Hyperbolic covariance geometry and the Szegő kernel

5.1 Covariance and correlation

The mixed cumulant formula at total order two gives

$$K_X(q, r) := \text{Cov}(X_q, X_r) = \frac{(1 - q)(1 - r)}{12(1 - qr)}. \quad (30)$$

After division by (3), the correlation kernel is

$$R(q, r) = \frac{\sqrt{(1 - q^2)(1 - r^2)}}{1 - qr}. \quad (31)$$

Introduce hyperbolic coordinates

$$q = \tanh u, \quad u = \text{artanh } q. \quad (32)$$

The subtraction identities immediately give

$$\frac{r-q}{1-qr} = \tanh(v-u), \quad R(q, r) = \operatorname{sech}(v-u). \quad (33)$$

Consequently

$$\operatorname{Vol} Z_Q = \prod_{i < j} \tanh(u_j - u_i). \quad (34)$$

The support volume depends only on hyperbolic separations, not on a common translation of all u_i .

5.2 Cauchy determinant and support–correlation duality

Let $R_Q = [R(q_i, q_j)]_{i,j=1}^d$. Since

$$R_Q = D \left[\frac{1}{1 - q_i q_j} \right] D, \quad D = \operatorname{diag} \sqrt{1 - q_i^2},$$

the Cauchy determinant gives

$$\det R_Q = \prod_{i < j} \left(\frac{q_j - q_i}{1 - q_i q_j} \right)^2. \quad (35)$$

Comparison with (24) proves the central identity.

Theorem 5.1 (Support–correlation identity). *For every ordered positive parameter set Q ,*

$$\boxed{\operatorname{Vol}_d Z_Q = \sqrt{\det R_Q}}. \quad (36)$$

In dimension two, if $\Delta u = v - u$, the support area A and correlation ρ satisfy

$$A = \tanh \Delta u, \quad \rho = \operatorname{sech} \Delta u, \quad A^2 + \rho^2 = 1. \quad (37)$$

Remark 5.2. Equation (36) is not a generic zonoid identity. Usually the volume of a Minkowski sum is an ℓ^1 sum of maximal minors, whereas a Gram determinant is an ℓ^2 sum by Cauchy–Binet. Here positivity of every generalized Vandermonde minor and the Littlewood identity collapse the former sum to the square root of the latter determinant.

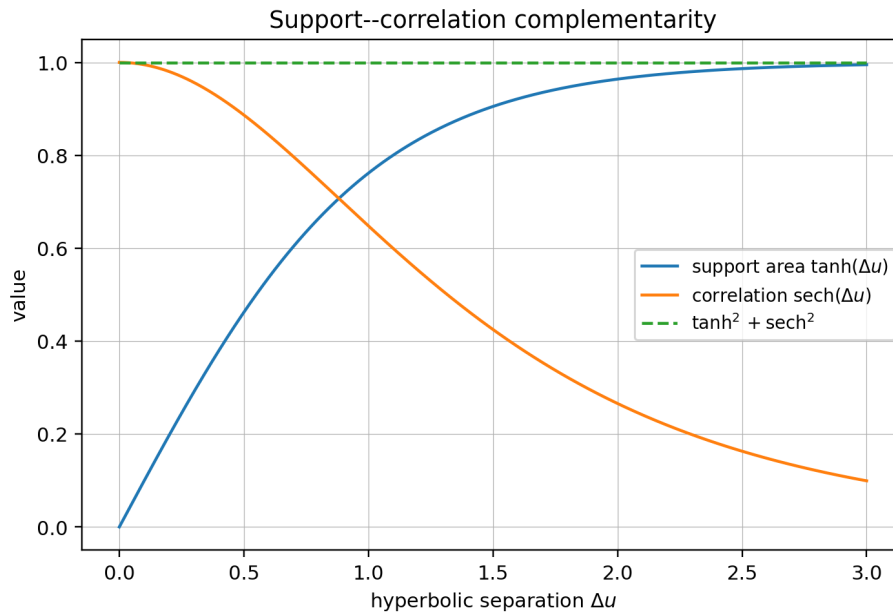


Figure 1: In dimension two, hyperbolic separation simultaneously increases support area and decreases correlation, with the exact Pythagorean relation (37).

5.3 Exact linear prediction errors

Define the standardized process

$$Y_q = \frac{X_q - 1/2}{\sqrt{(1-q)/(12(1+q))}}. \quad (38)$$

For distinct nodes $q_1, \dots, q_m$ and a further q , let

$$\hat{Y}_q = \operatorname{argmin}_{Z \in \operatorname{span}(Y_{q_1}, \dots, Y_{q_m})} \mathbb{E}(Y_q - Z)^2$$

denote the best linear predictor in L^2 .

Theorem 5.3 (Pseudohyperbolic prediction error). *One has*

$$\inf_{\beta \in \mathbb{R}^m} \mathbb{E} \left(Y_q - \sum_{j=1}^m \beta_j Y_{q_j} \right)^2 = \prod_{j=1}^m \left(\frac{q - q_j}{1 - qq_j} \right)^2. \quad (39)$$

For $q_1 < \dots < q_d$, the successive innovation standard deviations are

$$\sigma_k^{\text{lin}} = \prod_{i < k} \frac{q_k - q_i}{1 - q_i q_k}, \quad 2 \leq k \leq d, \quad (40)$$

and

$$\operatorname{Vol} Z_Q = \prod_{k=2}^d \sigma_k^{\text{lin}}. \quad (41)$$

Proof. The squared residual is the Schur complement of the node correlation matrix in the augmented correlation matrix containing q . It is therefore the ratio of the two determinants. Formula (35) cancels all old pair factors and leaves (39). Applying this sequentially gives (40); their product is (24). $\square$

The identity has a useful interpretation: the deterministic volume of the entire support equals the product of purely second-order innovation scales. It turns a nonlinear convex-geometric quantity into a sequence of exact linear-prediction errors.

5.4 Hardy-space feature map and strict total positivity

Let

$$\epsilon_n = \sqrt{12} (U_n - 1/2), \quad \mathbb{E}\epsilon_n = 0, \quad \mathbb{E}\epsilon_n^2 = 1. \quad (42)$$

Then

$$Y_q = \sum_{n \geq 0} \sqrt{1 - q^2} q^n \epsilon_n. \quad (43)$$

The coefficient vector

$$h_q = \sqrt{1 - q^2} (1, q, q^2, \dots) \in \ell^2 \quad (44)$$

is the normalized real Szegő kernel vector of the Hardy space $H^2(\mathbb{D})$ [14]. Hence $R(q, r) = \langle h_q, h_r \rangle$, and (39) is the classical normalized-kernel distance formula expressed probabilistically.

More generally, set

$$G(z) = \sum_{n \geq 0} \epsilon_n z^n, \quad z \in \mathbb{D}. \quad (45)$$

Although G is not Gaussian, its covariance kernel is exactly

$$\mathbb{E}[G(z)\overline{G(w)}] = \frac{1}{1 - z\bar{w}}. \quad (46)$$

For complex nodes $z_1, \dots, z_d$, the normalized Gram determinant is

$$\det \left[\frac{\sqrt{1-|z_i|^2} \sqrt{1-|z_j|^2}}{1-z_i \bar{z}_j} \right] = \prod_{i < j} \left| \frac{z_j - z_i}{1 - z_i \bar{z}_j} \right|^2. \quad (47)$$

Thus the real support volume is the square root of a finite Blaschke/Pick determinant.

The rectangular Cauchy formula also gives, for increasing real sequences $u_1 < \dots < u_d$ and $v_1 < \dots < v_d$,

$$\det[\operatorname{sech}(u_i - v_j)]_{i,j=1}^d = \frac{\prod_i \operatorname{sech} u_i \prod_j \operatorname{sech} v_j \prod_{i < j} (\tanh u_j - \tanh u_i) \prod_{i < j} (\tanh v_j - \tanh v_i)}{\prod_{i,j} (1 - \tanh u_i \tanh v_j)} > 0. \quad (48)$$

Hence the kernel $\operatorname{sech}(u - v)$ is strictly totally positive. The source repository already treats total-positivity themes; the contribution here is the explicit determinant arising from the common-digit covariance.

6 The exact two-parameter support lens

Fix $0 < q < r < 1$ and write $Z_{q,r}$ for the support of (X_q, X_r) . The generator slopes

$$\frac{(1-r)r^n}{(1-q)q^n} = \frac{1-r}{1-q} \left(\frac{r}{q} \right)^n \quad (49)$$

are strictly increasing. Consequently the boundary of every finite truncation is obtained by adding generators in slope order, and the infinite boundary has an exact geometric-comb description.

6.1 Comb vertices and interpolation

Set

$$\alpha = \frac{\log r}{\log q} \in (0, 1). \quad (50)$$

Let $g_{q,r} : [0, 1] \rightarrow [0, 1]$ be the continuous piecewise-linear interpolant through

$$g_{q,r}(q^m) = r^m = (q^m)^\alpha, \quad m = 0, 1, 2, \dots, \quad g_{q,r}(0) = 0. \quad (51)$$

Theorem 6.1 (Geometric-comb boundary). *The two-dimensional support is*

$$Z_{q,r} = \{(x, y) \in [0, 1]^2 : 1 - g_{q,r}(1 - x) \leq y \leq g_{q,r}(x)\}. \quad (52)$$

Its upper boundary vertices are (q^m, r^m) and its lower boundary vertices are their central reflections $(1 - q^m, 1 - r^m)$.

Proof. The tail of the generator sequence beginning at m sums to

$$\sum_{n \geq m} ((1-q)q^n, (1-r)r^n) = (q^m, r^m).$$

These tail sums are the successive vertices of the boundary obtained by ordering segments by slope. The opposite chain is its reflection through $(1/2, 1/2)$. Convexity fills exactly the region between the two polygonal limits, yielding (52). $\square$

6.2 Exact log-periodic modulation

For $x = q^{m+\theta}$ with $m \in \mathbb{N}_0$ and $0 \leq \theta \leq 1$, linear interpolation gives

$$g_{q,r}(x) = r^m \left[r + \frac{1-r}{1-q} (q^\theta - q) \right] = x^\alpha \Pi_{q,r}(\theta), \quad (53)$$

where

$$\Pi_{q,r}(\theta) = \frac{r + \frac{1-r}{1-q} (q^\theta - q)}{r^\theta}, \quad \Pi_{q,r}(0) = \Pi_{q,r}(1) = 1. \quad (54)$$

The fractional part of $\log x / \log q$ is exactly θ , so (53) is a multiplicatively periodic modulation. Since $0 < \alpha < 1$, x^α is concave and every chord lies below it; therefore

$$0 < \Pi_{q,r}(\theta) \leq 1. \quad (55)$$

This is an exact discrete-scale oscillation, not merely an endpoint asymptotic.

Integrating the trapezoids gives

$$\int_0^1 g_{q,r}(x) dx = \frac{1}{2} \sum_{m \geq 0} (q^m - q^{m+1})(r^m + r^{m+1}) \quad (56)$$

$$= \frac{(1-q)(1+r)}{2(1-qr)}. \quad (57)$$

By symmetry, the lens area is $2 \int_0^1 g - 1$, hence

$$\text{Area}(Z_{q,r}) = \frac{r-q}{1-qr}, \quad (58)$$

which independently recovers the $d = 2$ case of [theorem 4.1](#).

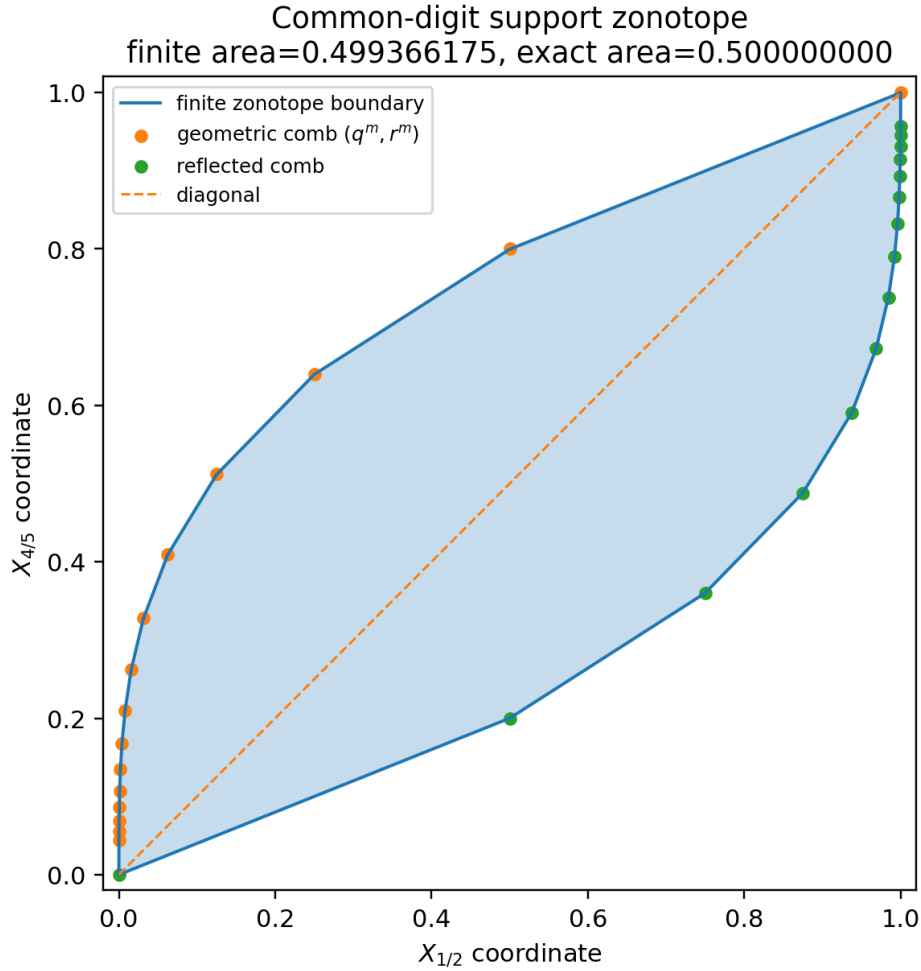


Figure 2: The common-digit support for $(q, r) = (1/2, 4/5)$. The upper and lower boundaries pass through the geometric comb and its reflection. The exact area is $1/2$.

Remark 6.2 (Fabius geometric-comb lens). At $q = 1/2$, the horizontal marginal CDF is the Fabius function. Thus (52) supplies an exact convex support geometry whose first coordinate is Fabius-distributed and whose boundary is a dyadic comb. This support description is later transported through the inverse Fabius map in section 10.

7 The hyperbolic-secant process and Gaussianization

7.1 A wide-sense stationary process

Using (42), put $q = \tanh u$ in (43) and write

$$Y_u := Y_{\tanh u} = \sum_{n \geq 0} \operatorname{sech} u \tanh^n u \epsilon_n, \quad u > 0. \quad (59)$$

Then

$$\mathbb{E}Y_u = 0, \quad \mathbb{E}(Y_u Y_v) = \operatorname{sech}(u - v). \quad (60)$$

Thus Y_u is stationary at second order. It is not strictly stationary: for example,

$$\kappa_4(Y_u) = -\frac{6}{5} \frac{1 - \tanh^2 u}{1 + \tanh^2 u}, \quad (61)$$

which varies with u .

With the convention

$$K(h) = \int_{\mathbb{R}} e^{i\omega h} s(\omega) d\omega,$$

the Fourier transform of $\operatorname{sech} h$ gives the spectral density

$$s(\omega) = \frac{1}{2} \operatorname{sech}\left(\frac{\pi\omega}{2}\right). \quad (62)$$

This is the centered hyperbolic-secant probability density.

7.2 Exact mixed cumulants under hyperbolic translation

Let $m \geq 2$. The standardized digit has cumulants

$$\kappa_m(\epsilon_0) = \begin{cases} 12^{m/2} B_m/m, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases} \quad (63)$$

The common coefficient sequence gives an exact multitime formula.

Theorem 7.1 (Hyperbolic mixed cumulants). *For $u_1, \dots, u_m > 0$,*

$$\kappa(Y_{u_1}, \dots, Y_{u_m}) = \kappa_m(\epsilon_0) \frac{\prod_{j=1}^m \operatorname{sech} u_j}{1 - \prod_{j=1}^m \tanh u_j}. \quad (64)$$

Let $u_j = T + t_j$, put $a_j = e^{-2t_j}$ and $z = e^{-2T}$. Then the same cumulant has the exact finite symmetric-polynomial form

$$\kappa(Y_{T+t_1}, \dots, Y_{T+t_m}) = \kappa_m(\epsilon_0) \frac{2^{m-1} z^{m/2-1} \prod_j a_j^{1/2}}{e_1(a) + e_3(a)z^2 + e_5(a)z^4 + \dots}, \quad (65)$$

where the denominator stops at the largest odd degree not exceeding m .

Proof. Apply cumulant additivity to (59) and sum the geometric series in $\prod_j \tanh u_j$, proving (64). Next use

$$\tanh(T + t_j) = \frac{1 - za_j}{1 + za_j}, \quad \operatorname{sech}(T + t_j) = \frac{2z^{1/2}a_j^{1/2}}{1 + za_j}.$$

The denominator becomes

$$\prod_j (1 + za_j) - \prod_j (1 - za_j) = 2 \sum_{\ell \text{ odd}} e_\ell(a) z^\ell,$$

and cancellation of $\prod_j (1 + za_j)$ yields (65). $\square$

In particular, uniformly for $(t_1, \dots, t_m)$ in compact sets,

$$\kappa(Y_{T+t_1}, \dots, Y_{T+t_m}) = \kappa_m(\epsilon_0) \frac{2^{m-1} e^{-(m-2)T - \sum_j t_j}}{\sum_j e^{-2t_j}} (1 + O(e^{-4T})). \quad (66)$$

For one time this reduces to

$$\kappa_m(Y_q) = \kappa_m(\epsilon_0) \frac{(1 - q^2)^{m/2}}{1 - q^m}. \quad (67)$$

All higher cumulants vanish as $q \uparrow 1$, at an explicitly graded rate.

7.3 Gaussian limit in smooth path spaces

Theorem 7.2 (Gaussianization by hyperbolic translation). *For $T > 0$, define the shifted process on a fixed compact interval $I \subset \mathbb{R}$ by*

$$Z_T(t) = Y_{T+t}, \quad t \in I. \quad (68)$$

For every finite $k \geq 0$, as $T \rightarrow \infty$,

$$Z_T \Longrightarrow G \quad \text{in } C^k(I), \quad (69)$$

where G is the centered stationary Gaussian process with covariance

$$\mathbb{E}(G(s)G(t)) = \operatorname{sech}(s - t). \quad (70)$$

Proof. For finitely many evaluation points and derivatives, Z_T is a triangular-array sum of the bounded independent variables ϵ_n . Differentiating the coefficient $c_n(u) = \operatorname{sech} u \tanh^n u$ a fixed number of times produces $c_n(u)$ times a polynomial in $n \operatorname{sech}^2 u / \tanh u$ and bounded hyperbolic functions. Uniformly for $t \in I$, its largest absolute coefficient is $O_k(e^{-T})$: the elementary bound $n^a e^{-cne^{-2T}} = O_a(e^{2aT})$ is exactly canceled by the derivative factors e^{-2aT} , leaving the leading $\operatorname{sech}(T + t)$.

The covariance matrices are independent of T and are obtained by differentiating $\operatorname{sech}(s - t)$. The maximal coefficient tends to zero, so the multivariate Lindeberg theorem proves joint Gaussian convergence for every finite derivative-evaluation family [17]. Tightness follows from the exact derivative variance bounds

$$\mathbb{E} \left| Y_u^{(a)} \right|^2 = |E_{2a}| \quad (71)$$

proved in section 8, and the L^2 increment estimate

$$\begin{aligned} \mathbb{E} \left| Y_{u+h}^{(a)} - Y_u^{(a)} \right|^2 &\leq |h|^2 \sup_v \mathbb{E} \left| Y_v^{(a+1)} \right|^2. \end{aligned}$$

Kolmogorov's criterion applied successively to derivatives gives tightness in $C^k(I)$. Finite-dimensional convergence and tightness complete the proof. $\square$

Remark 7.3. The covariance is already stationary for every T ; translation does not change any second-order statistic. It removes only the non-Gaussian higher-order structure. The common-digit curve therefore supplies a concrete nonstationary-to-stationary Gaussianization mechanism with an invariant covariance kernel.

7.4 Bell partitions beyond Wick's rule

Let $2p$ time points $t_1, \dots, t_{2p}$ be fixed. The moment-cumulant formula and (66) show that pair partitions survive at order one, a single four-block contributes at order e^{-2T} , and every remaining partition contributes $O(e^{-4T})$. If $\mathfrak{M}(S)$ denotes pairings of a finite even set S and

$$C_4(t_i, t_j, t_k, t_\ell) = -\frac{48}{5} \frac{e^{-(t_i+t_j+t_k+t_\ell)}}{e^{-2t_i} + e^{-2t_j} + e^{-2t_k} + e^{-2t_\ell}}, \quad (72)$$

then

$$\mathbb{E} \prod_{a=1}^{2p} Y_{T+t_a} = \sum_{M \in \mathfrak{M}([2p])} \prod_{\{i,j\} \in M} \operatorname{sech}(t_i - t_j)$$

$$+ e^{-2T} \sum_{\substack{B \subset [2p] \\ |B|=4}} C_4(t_B) \sum_{M \in \mathfrak{M}([2p] \setminus B)} \prod_{\{i,j\} \in M} \operatorname{sech}(t_i - t_j) + O(e^{-4T}). \quad (73)$$

The error is uniform on compact time sets. This formula is an explicit Bell-polynomial deformation of Wick's theorem. Higher corrections are generated mechanically by blocks of sizes 4, 6, 8, $\dots$ using (65).

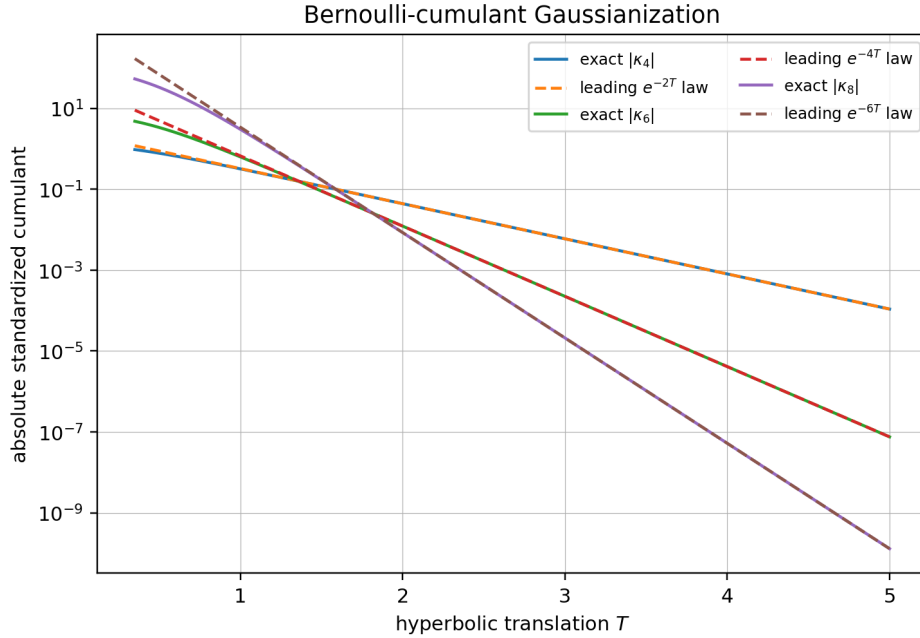


Figure 3: Exact one-point cumulants of orders 4, 6, 8 and their leading Bernoulli asymptotics. The slopes are $-(m - 2)$ in hyperbolic time.

8 Euler determinants and a local derivative-innovation ladder

The stationary covariance (60) contains more arithmetic structure than is visible from correlation alone. Let the Euler secant numbers be defined by

$$\operatorname{sech} z = \sum_{m=0}^{\infty} E_{2m} \frac{z^{2m}}{(2m)!}, \quad E_{2m} = (-1)^m |E_{2m}|. \quad (74)$$

Differentiating the covariance gives

$$\mathbb{E}[Y_u^{(a)} Y_v^{(b)}] = (-1)^b \operatorname{sech}^{(a+b)}(u - v). \quad (75)$$

At one point,

$$\operatorname{Var}(Y_u^{(m)}) = |E_{2m}|, \quad (76)$$

which justifies (71).

8.1 A confluent determinant equal to one

Define the $d \times d$ normalized derivative covariance matrix

$$C_{ij}^{(d)} = \frac{(-1)^j \operatorname{sech}^{(i+j)}(0)}{i!j!}, \quad 0 \leq i, j \leq d-1. \quad (77)$$

Its first few leading blocks are

$$(1), \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & 5/4 \end{pmatrix}.$$

Theorem 8.1 (Unit derivative-jet determinant). *For every $d \geq 1$,*

$$\boxed{\det C^{(d)} = 1.} \quad (78)$$

Equivalently, if

$$\mu_{2m} = |E_{2m}|, \quad \mu_{2m+1} = 0, \quad (79)$$

then

$$\boxed{\det[\mu_{i+j}]_{i,j=0}^{d-1} = \prod_{k=0}^{d-1} (k!)^2.} \quad (80)$$

Proof. Choose distinct nodes $u_1, \dots, u_d$ tending to a common u . Apply the Newton divided-difference transform to the random vector $(Y_{u_1}, \dots, Y_{u_d})$. The determinant of this transform is the reciprocal of the Vandermonde $\prod_{i < j} (u_j - u_i)$. The transformed covariance determinant is therefore

$$\frac{\det[\operatorname{sech}(u_i - u_j)]}{\prod_{i < j} (u_j - u_i)^2} = \prod_{i < j} \left(\frac{\tanh(u_j - u_i)}{u_j - u_i} \right)^2, \quad (81)$$

where (35) was used in hyperbolic coordinates. The right side tends to one. Divided differences converge to $Y_u^{(k)}/k!$, so the left side tends to $\det C^{(d)}$, proving (78).

The spectral density (62) has moments (79). The derivative covariance matrix differs from the raw moment matrix only by row and column phase factors and the factorial normalizations. Undoing these normalizations multiplies the determinant by $\prod_{k < d} (k!)^2$, proving (80). $\square$

8.2 Meixner–Pollaczek polynomials and exact innovations

Let p_n be the monic orthogonal polynomial sequence for the probability density $s(\omega)$ in (62). Rescaling the symmetric Meixner–Pollaczek polynomials with parameters $\lambda = 1/2$ and $\phi = \pi/2$ [15] gives

$$p_0(\omega) = 1, \quad p_1(\omega) = \omega, \quad p_{n+1}(\omega) = \omega p_n(\omega) - n^2 p_{n-1}(\omega). \quad (82)$$

Their norms are

$$\int_{\mathbb{R}} p_m(\omega) p_n(\omega) s(\omega) d\omega = (n!)^2 \delta_{mn}. \quad (83)$$

The norm product in (83) supplies a second proof of the Hankel determinant (80).

Define real differential-operator polynomials $Q_n(D)$ by

$$Q_0(D) = 1, \quad Q_1(D) = D, \quad Q_{n+1}(D) = DQ_n(D) + n^2 Q_{n-1}(D). \quad (84)$$

Equivalently, $Q_n(D) = i^n p_n(-iD)$. The first operators are

$$Q_0 = 1, \quad Q_1 = D, \quad Q_2 = D^2 + 1, \quad (85)$$

$$Q_3 = D^3 + 5D, \quad Q_4 = D^4 + 14D^2 + 9, \quad Q_5 = D^5 + 30D^3 + 89D. \quad (86)$$

Theorem 8.2 (Local derivative innovations). *For every fixed u , define*

$$Z_n(u) = \frac{1}{n!} Q_n(D) Y_u. \quad (87)$$

Then

$$\mathbb{E}[Z_m(u) Z_n(u)] = \delta_{mn}. \quad (88)$$

Thus each successive normalized derivative jet has linear innovation variance one. The variables are generally not independent, because the underlying process is non-Gaussian at finite u .

Proof. In the spectral representation, $Q_n(D)$ has multiplier $i^n p_n(\omega)$. Formula (83) gives zero covariance for unequal degrees and variance $(n!)^2$ for degree n . Division by $n!$ proves (88). $\square$

Proved in this report

The recurrence (84) gives an exact, local, differential analogue of Gram–Schmidt orthogonalization. For example,

$$Z_2 = (Y'' + Y)/2, \quad Z_3 = (Y''' + 5Y')/6$$

are unit-variance and orthogonal to all earlier Z_j at the same point. The coefficients 1, 5, 14, 9, 30, 89, ... are forced by the hyperbolic-secant spectral law.

9 Confluent parameter jets and exact support volumes

9.1 Samplewise analyticity and jet supports

Recall

$$a_n(q) = (1 - q)q^n = q^n - q^{n+1}. \quad (89)$$

For each digit sequence, $X_q = \sum_n a_n(q) U_n$ is analytic on $|q| < 1$. For $d \geq 1$ define the divided-derivative jet

$$J_d(q) = \left(X_q, X'_q, \frac{X''_q}{2!}, \dots, \frac{X_q^{(d-1)}}{(d-1)!} \right). \quad (90)$$

Its support is the infinite zonotope generated by

$$w_n^{(d)}(q) = \left(a_n(q), a'_n(q), \dots, \frac{a_n^{(d-1)}(q)}{(d-1)!} \right)^\top. \quad (91)$$

9.2 Confluence of the exact volume formula

For distinct nodes $q_1, \dots, q_d$, let T_Q be the Newton divided-difference transform

$$(f(q_1), \dots, f(q_d)) \mapsto (f[q_1], f[q_1, q_2], \dots, f[q_1, \dots, q_d]). \quad (92)$$

Its determinant is

$$|\det T_Q| = \frac{1}{\prod_{i < j} (q_j - q_i)}. \quad (93)$$

As all nodes tend to q , the transformed common-digit vector tends samplewise and in support to $J_d(q)$.

Theorem 9.1 (Exact parameter-jet support volume). *For $0 < q < 1$,*

$$\boxed{\text{Vol}_d \text{supp } J_d(q) = (1 - q^2)^{-\binom{d}{2}}.} \quad (94)$$

Proof. Apply T_Q to the support Z_Q . Linear transformation multiplies volume by (93); hence theorem 4.1 gives

$$\text{Vol}(T_Q Z_Q) = \frac{1}{\prod_{i < j} (1 - q_i q_j)}. \quad (95)$$

The transformed generators converge absolutely to (91) as $q_i \rightarrow q$, so the convex bodies converge in Hausdorff distance. Volume continuity and (95) give (94). $\square$

Remark 9.2 (Why positivity is essential). The samplewise jet $J_d(q)$ exists throughout $|q| < 1$, but the product formula (94) is a positive-parameter statement. For $q < 0$ the maximal minors of the generator matrix change orientation, so the zonotope volume is no longer obtained by the unsigned Schur–Littlewood sum used above. The signed regime is separated into theorem 14.1 rather than silently extending (94) beyond its valid chamber.

Example 9.3. At the Fabius parameter $q = 1/2$,

$$\text{Vol supp } J_d(1/2) = \left(\frac{4}{3}\right)^{\binom{d}{2}}. \quad (96)$$

For $d = 2, 3, 4$ these values are $4/3$, $64/27$, and $4096/729$.

9.3 Several confluent clusters

Let $0 < p_1 < \dots < p_s < 1$ be distinct points with multiplicities $m_1, \dots, m_s$. Consider the concatenated local jets

$$\mathcal{J} = \bigoplus_{a=1}^s J_{m_a}(p_a). \quad (97)$$

Its dimension is $d = \sum_a m_a$.

Theorem 9.4 (Multi-jet support volume). *The support of (97) has volume*

$$\boxed{\text{Vol supp } \mathcal{J} = \frac{\prod_{1 \leq a < b \leq s} \left(\frac{p_b - p_a}{1 - p_a p_b} \right)^{m_a m_b}}{\prod_{a=1}^s (1 - p_a^2)^{\binom{m_a}{2}}}.} \quad (98)$$

Proof. Start with m_a distinct nodes tending to each p_a and apply divided-difference transforms separately inside each cluster. The within-cluster Vandermonde factors cancel from (24); cross-cluster factors converge to the numerator in (98), while within-cluster denominators converge to the displayed powers of $1 - p_a^2$. $\square$

9.4 Hyperbolic collision expansion

If $u_i = u + \varepsilon s_i$, then (34) and $\tanh x = x - x^3/3 + O(x^5)$ give the even expansion

$$\text{Vol } Z = \varepsilon^{\binom{d}{2}} \prod_{i < j} (s_j - s_i) \left[1 - \frac{\varepsilon^2}{3} \sum_{i < j} (s_j - s_i)^2 + O(\varepsilon^4) \right]. \quad (99)$$

Unlike the corresponding expansion in raw q -coordinates, no first-order translation term appears. This is another indication that $u = \text{artanh } q$ is the intrinsic parameter.

9.5 Exact jet generators and Lambert- W cutoff scales

Differentiating (89) gives the exact formula

$$\frac{a_n^{(k)}(q)}{k!} = \binom{n}{k} q^{n-k} - \binom{n+1}{k} q^{n+1-k}, \quad (100)$$

with the usual convention that $\binom{n}{k} = 0$ for $n < k$. For fixed k and $0 < q < 1$,

$$\frac{a_n^{(k)}(q)}{k!} \sim \frac{1-q}{k!} n^k q^{n-k}. \quad (101)$$

Thus a truncation threshold reduces to solving $n^k q^n = \varepsilon$. If $\lambda = -\log q$, the large solution is

$$n_\varepsilon = -\frac{k}{\lambda} W_{-1} \left(-\frac{\lambda}{k} \varepsilon^{1/k} \right), \quad (102)$$

where W_{-1} is the negative real branch of Lambert's function [16]. Constants from (101) are absorbed by replacing ε accordingly. Formula (102) gives a principled adaptive truncation rule for jet-volume and jet-density computations.

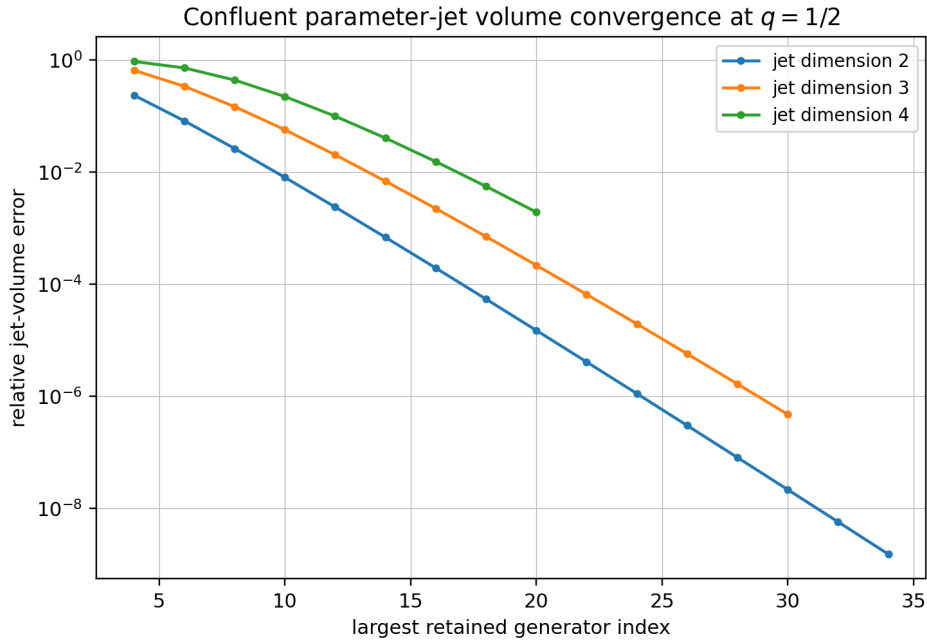


Figure 4: Convergence of finite-generator jet volumes to (94) at $q = 1/2$. Higher jets require more generators because polynomial factors n^k delay the geometric tail.

10 Legendre coefficients, copulas, and the inverse Fabius function

10.1 A bivariate shifted-Legendre calculus

Let $f_{q,r}$ be the C_c^∞ density of (X_q, X_r) and define shifted Legendre polynomials

$$\mathcal{P}_m(x) = P_m(2x - 1). \quad (103)$$

They satisfy

$$\int_0^1 \mathcal{P}_m(x) \mathcal{P}_n(x) dx = \frac{\delta_{mn}}{2m+1}.$$

Therefore

$$f_{q,r}(x, y) = \sum_{m,n \geq 0} (2m+1)(2n+1) L_{mn}(q, r) \mathcal{P}_m(x) \mathcal{P}_n(y), \quad (104)$$

where

$$L_{mn}(q, r) = \mathbb{E}[\mathcal{P}_m(X_q) \mathcal{P}_n(X_r)]. \quad (105)$$

Every coefficient is an exact rational function of q, r : expand the two polynomials and evaluate each mixed monomial by (16). Central symmetry implies

$$L_{mn}(q, r) = 0 \quad \text{when } m+n \text{ is odd.} \quad (106)$$

Since the zero extension of $f_{q,r}$ is C^∞ on $\mathbb{R}^2$, repeated integration by parts with the Legendre Sturm–Liouville operator gives, for every $A > 0$,

$$|L_{mn}(q, r)| \leq C_A (1+m+n)^{-A}. \quad (107)$$

For compact formulas, put

$$v_q = \frac{1-q}{12(1+q)}, \quad v_r = \frac{1-r}{12(1+r)}, \quad c = \frac{(1-q)(1-r)}{12(1-qr)}, \quad (108)$$

and

$$k_{22} = -\frac{1}{120} \frac{(1-q)^2(1-r)^2}{1-q^2r^2}. \quad (109)$$

Then

$$L_{00} = 1, \quad L_{11} = 4c, \quad (110)$$

$$L_{20} = 6v_q - \frac{1}{2} = -\frac{q}{1+q}, \quad L_{02} = 6v_r - \frac{1}{2} = -\frac{r}{1+r}, \quad (111)$$

$$L_{22} = 36(v_q v_r + 2c^2 + k_{22}) - 3(v_q + v_r) + \frac{1}{4}. \quad (112)$$

At $(q, r) = (1/2, 4/5)$,

(m, n)	L_{mn}	$(\mathbf{m}, \mathbf{n})$	L_{mn}
$(0, 0)$	1	$(1, 1)$	1/18
$(2, 0)$	-1/3	$(0, 2)$	-4/9
$(2, 2)$	599/3780	$(1, 2), (2, 1)$	0

The accompanying Python script derives these values symbolically and also emits the fully expanded rational expression for L_{22} .

10.2 The common-digit copula

Let $Q_q = F_q^{-1}$ denote the marginal quantile. On the open unit square, the common-digit copula has density

$$c_{q,r}(u, v) = \frac{f_{q,r}(Q_q(u), Q_r(v))}{f_q(Q_q(u)) f_r(Q_r(v))}. \quad (113)$$

This is formally standard, but the exact support lens makes the transformed support explicit. From (52), define

$$v_+(u) = F_r(g_{q,r}(Q_q(u))), \quad (114)$$

$$v_-(u) = F_r(1 - g_{q,r}(1 - Q_q(u))). \quad (115)$$

Then

$$\text{supp}(c_{q,r}) = \{(u, v) \in [0, 1]^2 : v_-(u) \leq v \leq v_+(u)\}. \quad (116)$$

At $q = 1/2$, $Q_q = F^{-1}$ is the inverse Fabius function. Thus

$$v_+(u) = F_r(g_{1/2,r}(F^{-1}(u))), \quad v_-(u) = F_r(1 - g_{1/2,r}(1 - F^{-1}(u))). \quad (117)$$

Equation (117) is a direct, exact bridge from inverse-Fabius asymptotics to a genuinely bivariate dependence geometry. Endpoint expansions for these curves are proposed in [theorem 14.5](#).

11 A Thue–Morse vector filter

Let

$$\tau_n = (-1)^{\text{wt}_2(n)}, \quad (118)$$

where $\text{wt}_2(n)$ is the binary digit sum. The finite product identity

$$\sum_{n=0}^{2^m-1} \tau_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}) \quad (119)$$

belongs to the established Thue–Morse input of the repository. Applying it coordinatewise to the common-digit generators gives a useful new vector packaging.

Proposition 11.1 (Thue–Morse block vector). *For v_n from (9),*

$$\sum_{n=0}^{2^m-1} \tau_n v_n = \left((1 - q_i) \prod_{j=0}^{m-1} (1 - q_i^{2^j}) \right)_{i=1}^d. \quad (120)$$

More generally, for every jet order k ,

$$\sum_{n=0}^{2^m-1} \tau_n \frac{a_n^{(k)}(q)}{k!} = \frac{1}{k!} \frac{d^k}{dq^k} \left[(1 - q) \prod_{j=0}^{m-1} (1 - q^{2^j}) \right]. \quad (121)$$

Proof. Multiply (119) by $1 - q_i$ and evaluate at each q_i . Differentiation gives (121). $\square$

The right side of (121) has a zero of order $m + 1$ at $q = 1$. Hence the filter annihilates the first m endpoint jet orders. This is a continuous-parameter analogue of the Prouhet cancellation

$$\sum_{n < 2^m} \tau_n n^k = 0, \quad 0 \leq k < m. \quad (122)$$

A promising next step is to use several block levels as alternative generators for the support zonoid and analyze the resulting determinants through products (120); see [theorem 14.8](#).

12 Hyperbolic node design

Because $\text{Vol } Z_Q^2 = \det R_Q$, choosing parameters to maximize support volume is exactly the D -optimal design problem for the sech correlation kernel. On a bounded hyperbolic interval, this problem has a clean variational structure.

Fix $L > 0$ and consider

$$\Psi(u_1, \dots, u_d) = \sum_{i < j} \log \tanh(u_j - u_i), \quad 0 \leq u_1 < \dots < u_d \leq L. \quad (123)$$

Theorem 12.1 (Unique volume-maximizing design). *There is a unique maximizer of (123). It satisfies*

$$u_1 = 0, \quad u_d = L, \quad (124)$$

and is reflection-symmetric,

$$u_k + u_{d+1-k} = L. \quad (125)$$

Every interior node satisfies the equilibrium equation

$$\sum_{j \neq k} \frac{2}{\sinh(2(u_k - u_j))} = 0, \quad 2 \leq k \leq d-1. \quad (126)$$

For $d = 3$, the optimizer is $(0, L/2, L)$.

Proof. The function $\log \tanh x$ is strictly concave on $(0, \infty)$ because

$$\frac{d}{dx} \log \tanh x = \frac{2}{\sinh 2x}, \quad \frac{d^2}{dx^2} \log \tanh x = -\frac{4 \cosh 2x}{\sinh^2 2x} < 0. \quad (127)$$

Thus Ψ is concave and is strictly concave modulo common translation. It tends to $-\infty$ at collisions. Translation invariance lets us place $u_1 = 0$; if $u_d < L$, dilating all gaps increases every pair distance and therefore increases Ψ , so $u_d = L$. Strict concavity on this affine slice gives uniqueness. Reflection about $L/2$ preserves the objective, so uniqueness forces (125). Differentiation gives (126). The $d = 3$ claim follows from symmetry. $\square$

The large- d limit is a nonclassical logarithmic-energy problem with pair interaction

$$-\log \tanh |u - v|.$$

Near the diagonal it has the usual $-\log |u - v|$ singularity, but its long-range interaction decays exponentially. Determining the equilibrium density and edge scaling is posed in [theorem 14.6](#).

13 Reproducible numerical experiments

All computations in this section are checks of exact formulas, not evidence substituted for proof. The archive contains `code/experiments.py`, with detailed comments, fixed random seed, exact SymPy calculations where possible, CSV outputs, and every figure in both PDF and PNG form.

13.1 Reproduction command and environment

From the archive root, run

```
python code/experiments.py --output-root .
```

The supplied run used Python 3.13, NumPy 2.3.5, SymPy 1.14.0, Matplotlib 3.10.8, seed 20260830, 200,000 Monte Carlo samples, and 80 geometric digits. Exact formulas do not depend on these numerical choices.

13.2 Finite-zonotope convergence

For $N \geq d-1$, the script sums all absolute maximal minors of $v_0, \dots, v_N$ and compares with (24). Representative terminal rows are:

Parameters	d	N	truncated volume	relative error
(0.5, 0.8)	2	40	0.499893661760792	2.13×10^{-4}
(0.2, 0.5, 0.8)	3	40	0.119012172968055	2.98×10^{-4}
(0.2, 0.4, 0.6, 0.8)	4	22	0.004049504148898	3.64×10^{-2}

The four-dimensional truncation converges more slowly because many maximal minors contain at least one late generator.

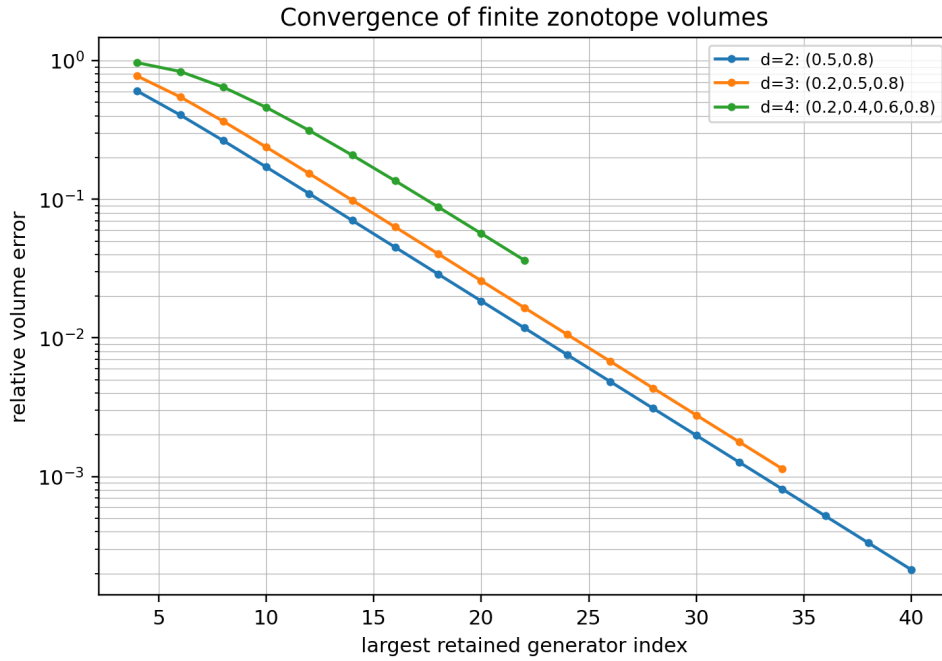


Figure 5: Relative error of finite zonotope volumes. The horizontal axis is the largest retained generator index.

13.3 Correlation determinants

For dimensions two through five, direct floating-point determinants of (31) agreed with the square of (24) to between 10^{-16} and 10^{-19} . For example,

Parameters	Vol Z	$\det R$	$\det R - (\text{Vol } Z)^2$
(0.5, 0.8)	0.5000000000000000	0.2500000000000000	5.6×10^{-17}
(0.15, 0.45, 0.75)	0.098489668345446	0.009700214770796	9.4×10^{-17}
(0.2, 0.4, 0.6, 0.8)	0.004202285801424	$1.76592059568538 \times 10^{-5}$	1.4×10^{-18}

13.4 Monte Carlo covariance

The same 80 uniforms were reused across $q = 0.2, 0.5, 0.8$, exactly as required by the common-digit coupling. With 200,000 samples,

(q, r)	empirical covariance	exact covariance	difference
(0.2, 0.5)	0.0371028472	0.0370370370	6.58×10^{-5}
(0.2, 0.8)	0.0158828618	0.0158730159	9.85×10^{-6}
(0.5, 0.8)	0.0138915783	0.0138888889	2.69×10^{-6}

The retained-digit covariance already agrees with the infinite formula to machine precision for these parameters; the visible differences are sampling error.

13.5 Jet volumes and Euler determinants

At $q = 1/2$, the truncated $d = 3$ jet volume with generators through $n = 32$ is

$$2.370370049098556,$$

versus the exact $64/27 = 2.370370370370370$, a relative error of 1.36×10^{-7} . For $d = 4$ and $n \leq 21$, the relative error is 1.13×10^{-3} . The symbolic Euler Hankel determinants were checked exactly through $d = 8$; every difference from $\prod_{k < d} (k!)^2$ was identically zero. The $d = 8$ determinant is

$$15728001190723584000000.$$

13.6 Gaussianization rates

The exact cumulants (67) were compared with their leading hyperbolic asymptotics. At $T = 3$, the ratios exact/leading are 0.99999386 for order four and 0.99995699 for order eight. This rapid agreement reflects the absence of an e^{-2T} correction inside the individual cumulant formula (65); after the leading factor, the first denominator correction is e^{-4T} .

Reproducible computational check

The numerical files preserve more digits than the rounded tables in the PDF. In particular:

- `volume_convergence.csv` and `jet_volume_convergence.csv` contain every truncation;
- `gram_volume_identity.csv` records correlation determinants;
- `euler_hankel_determinants.csv` contains exact integer checks;
- `gaussianization_cumulants.csv` contains exact and leading values;
- `monte_carlo_covariance.csv` records seed, sample size, finite-digit bias, and empirical error;
- `legendre_coefficients.tex` is generated symbolically.

14 Conjectures and promising research directions

The preceding results are exact. This section deliberately separates statements that remain unproved or only heuristically supported.

14.1 Signed parameters and oriented-matroid chambers

For $q \in (-1, 1)$, the analytic series (1) still makes sense, but its coefficients alternate when $q < 0$ and the support is no longer contained in the unit cube. The finite-zonotope volume remains a sum of absolute generalized Vandermonde minors, whose signs can change.

Conjecture 14.1 (Signed-volume chamber analyticity). *For distinct $q_1, \dots, q_d \in (-1, 1)$, the support volume is real-analytic on every connected component of the complement of the generalized-Vandermonde wall arrangement*

$$\det[q_i^{n_j}]_{i,j=1}^d = 0, \quad 0 \leq n_1 < \dots < n_d. \quad (128)$$

On each chamber it is represented by an absolutely convergent signed Schur series. Chambers on which all maximal minors have one orientation admit finite-product simplifications analogous to (24).

A first tractable case is $d = 2$ with one positive and one negative parameter, where the sign pattern reduces to the parity of exponent differences.

14.2 Complex parameters, Pick determinants, and real support volume

For $z \in \mathbb{D}$, define the complex geometric-uniform variable

$$X_z = (1 - z) \sum_{n \geq 0} z^n U_n. \quad (129)$$

A d -tuple has a real zonoid support in $\mathbb{C}^d \cong \mathbb{R}^{2d}$. Its covariance Gram matrix is still governed by the Szegő kernel, but real $2d$ -volume involves $2d \times 2d$ minors rather than the complex $d \times d$ determinant.

Conjecture 14.2 (Complex Pick-zonoid correspondence). *The intrinsic volumes of the complex common-digit support can be expressed through finite sums of principal minors of normalized Pick matrices and products of pseudohyperbolic distances. On conjugation-stable node sets, the top real volume should factor into a Pick determinant times an explicit phase/orientation correction.*

The natural language is exterior algebra over $\mathbb{R}$ applied to Hardy kernel vectors. The exact real identity (36) is the rank-one boundary case of this proposed theory.

14.3 Sharp ray-wise decay of the joint sinc product

Let $t = s\theta$ in (13) and let $q_*(\theta)$ be the largest q_j for which $\theta_j(1 - q_j) \neq 0$. Then

$$A_n(\theta) = \sum_j \theta_j(1 - q_j)q_j^n \sim c_*(\theta)q_*(\theta)^n.$$

The active factor count is approximately $\log s / |\log q_*|$.

Conjecture 14.3 (Generic joint sinc-product envelope). *Away from quantitatively controlled neighborhoods of the zero hyperplanes (14),*

$$\log |\Phi_Q(s\theta)| = -\frac{(\log s)^2}{2|\log q_*(\theta)|} + O_\theta(\log s), \quad s \rightarrow \infty. \quad (130)$$

When ratios $\log q_i / \log q_j$ are rational, the $O(\log s)$ term should contain a periodic function of $\log s$; irrational ratios should produce a quasiperiodic torus function. Directions lying on cancellation strata require replacement of q_ by the first surviving exponential scale.*

A proof should combine exponential-polynomial dominance with the one-parameter sine-product estimates already developed in the repository.

14.4 Jet small balls and Lambert phases

Consider a positive weighted sum with jet-size coefficients,

$$S_{m,q} = \sum_{n \geq 1} n^m q^n U_n, \quad \lambda = -\log q, \quad (131)$$

and put $L = \log(1/x)$. The active index obtained from (102) satisfies

$$\lambda N = L + (m + 1) \log L + O(1)$$

at the saddle relevant to small deviations: the extra $+1$ comes from the simplex/entropy factor in addition to the polynomial weight.

Conjecture 14.4 (Polynomial-geometric small-ball expansion). As $x \downarrow 0$,

$$\log \mathbb{P}(S_{m,q} \leq x) = -\frac{L^2}{2\lambda} - \frac{m+1}{\lambda} L \log L + A_{m,q} L + B_{m,q} (\log L)^2 + \mathcal{P}_{m,q} \left(\frac{L + (m+1) \log L}{\lambda} \right) + o(1). \quad (132)$$

where $\mathcal{P}_{m,q}$ is bounded and 1-periodic. The coefficients and phase should be recoverable from a Lambert- W_{-1} saddle and a Mellin analysis of the Laplace product.

Editorial note (ProveIt, 2026-09-29). As stated, Conjecture 14.4 is false whenever $\lambda \neq 1$, that is, whenever $q \neq e^{-1}$. The expansion of $\log \mathbb{P}(S_{m,q} \leq x)$ contains the unbounded term $\frac{(m+1)^2 \log \lambda}{\lambda} \log L$, which no choice of $A_{m,q}$, $B_{m,q}$ and bounded periodic $\mathcal{P}_{m,q}$ can absorb. With $p = m+1$ the correct form is $-\frac{L^2}{2\lambda} - \frac{p}{\lambda} L \log L + \left(\frac{1}{2} + \frac{p(1+\log \lambda)}{\lambda}\right) L - \frac{p^2}{2\lambda} (\log L - \log \lambda)^2 - \frac{p}{2} \log(2\pi) - \Psi_\lambda((L + p \log L - p \log \lambda)/\lambda) + O((\log L)^2/L)$, with an explicit one-periodic Ψ_λ ; for $\lambda = 1$ the conjectured shape holds. For the Fabius law ($m = 0$, $q = 1/2$) the term $\frac{\log \log 2}{\log 2} \log L$ is already part of the machine-checked small-argument expansion `Fabius.log_fabius_sub_explicitCorrectedWikipediaMain_isBigO` (Analysis/FabiusFunction/Lean/FabiusFunction/FabiusSharpAsymptotic.lean), whose elementary main term is `fabiusWikipediaElementaryMain`. The general correction, an all-order expansion in the exact coordinate $x = R^p e^{-\lambda R}$, and the case $\lambda = 1$ are in the unreviewed draft `Polynomial_Geometric_Small_Deviations_Fabius_Jets/` of this group (2026-09-29), Corollaries `cor:logform` and `cor:refutation`.

For $m = 0$, this is compatible with the log-quadratic and periodic endpoint structure of Fabius-type laws. The new problem is to track how polynomial jet weights deform every lower-order term.

14.5 Inverse-Fabius copula endpoint laws

Conjecture 14.5 (Copula boundary transseries). For fixed $r > 1/2$, the curves (117) admit endpoint transseries in which the inverse-Fabius Lambert- W scale is composed with the exact dyadic modulation $\Pi_{1/2,r}$. The leading logarithmic scale is inherited from F^{-1} , while the first nonconstant oscillation is the composition of the inverse-Fabius periodic phase with $\Pi_{1/2,r}$.

This should produce explicit asymptotics for conditional extreme events such as

$$\mathbb{P}(X_r \leq y \mid X_{1/2} = x)$$

in boundary layers where $x, y \downarrow 0$ together.

Editorial note (ProveIt, 2026-09-29). Joint lower-corner small-deviation rates of the common-digit family, their first correction, the standardized copula exponent, and conditional extremes given threshold events such as $\{X_{1/2} \leq x\}$ are proved in the unreviewed draft `Endpoint_Geometry_Common_Digit_Fabius_Laws/` of this group (2026-09-29), Theorems `thm:sharp`, `thm:copula` and `thm:diagonal` and display `eq:conditional-threshold`. Conjecture 14.5 and conditioning on an exact value $X_{1/2} = x$ remain open.

14.6 Hyperbolic Fekete limits

Question 14.6 (Large-design equilibrium). As $d \rightarrow \infty$ with $u_i \in [0, L]$ chosen by theorem 12.1, what is the limiting empirical measure? Does it have an explicit density solving

$$\int_0^L \log \tanh |u - v| d\mu(v) = \text{constant} \quad (133)$$

on its support, and what are the endpoint spacings?

The short-distance logarithmic singularity suggests an arcsine-like edge enhancement, while exponential screening at long distance suggests a flatter bulk than the classical logarithmic gas.

14.7 The q -Barnes constant

Question 14.7 (Asymptotics of the finite hyperbolic superfactorial). *Find a complete joint asymptotic for (26) when $d \rightarrow \infty$ and $h \downarrow 0$ with dh fixed or slowly varying. In particular, express $C(h)$ from (27) through q -Barnes/MacMahon functions and determine its modular transseries.*

This regime interpolates between coalescing parameter jets and high-dimensional separated designs.

14.8 Thue–Morse determinant filters

Question 14.8 (Filtered generator determinants). *Let $b_m(Q)$ be the vector on the right of (120). Determine closed forms, signs, and asymptotics of*

$$\det[b_{m_1}(Q) \cdots b_{m_d}(Q)]. \quad (134)$$

Can these vectors be used to give a multiscale decomposition of Z_Q whose block volumes factor into q -Pochhammer or Thue–Morse products?

Such a decomposition would link the present convex geometry directly to the repository’s discrete Thue–Morse exact-value machinery.

14.9 Legendre singular values and conditional inversion

The bivariate coefficients L_{mn} define a compact operator from the q marginal Legendre basis to the r basis. Its singular values quantify dependence beyond correlation.

Conjecture 14.9 (Superalgebraic dependence spectrum). *After conjugation by the square roots of the two marginal densities, the common-digit conditional expectation operator has singular values decaying faster than every power. Their sharp stretched-exponential scale is controlled by the closest complex singularities of the joint sinc product and changes at arithmetic resonances of $\log q / \log r$.*

At $q = 1/2$, the singular functions should encode inverse-Fabius composition in a way not visible from marginal Legendre series alone.

15 Formalization roadmap

The results divide naturally into Lean-sized modules.

- F1. Common-digit process.** Define $a_n(q)$, prove summability uniformly on compact q -sets, and construct X_q as an L^2 and almost-sure series. Existing geometric-series and Fabius infrastructure should cover most scalar lemmas.
- F2. Finite supports.** Represent $Z_Q^{(N)}$ as a linear image of a cube. Prove the Vandermonde full-rank statement and central symmetry.
- F3. Zonotope volume.** Import or prove the finite maximal-minor formula. The principal algebraic burden is the generalized Vandermonde–Schur factorization and Littlewood sum (23).
- F4. Cauchy determinant.** Formalize (35); this independently certifies the square of the proposed volume and provides a useful regression test for the Schur proof.
- F5. Hausdorff limits.** Show finite supports converge and invoke continuity of volume for convex bodies with nonempty interior.

- F6. Cumulants and moments.** First formalize the finite digit formulas, then pass to limits using absolute convergence. Bell partitions can reuse finite-set partition libraries.
- F7. Confluence.** Develop Newton divided differences as matrices, prove determinant (93), and pass to derivative jets. This yields both (94) and the Euler determinant (78).
- F8. Analytic/probabilistic layer.** Smooth density, Prékopa closure, association, and the process-level CLT can be staged after the exact algebraic core.

A productive first milestone is a formally verified chain

$$(19) \implies (24) \implies (35) \implies (36),$$

for finite rational parameter lists. It combines convex geometry, symmetric polynomials, and matrix algebra while remaining independent of delicate endpoint asymptotics.

16 Conclusion

Using one uniform digit sequence simultaneously at several geometric parameters exposes a structure that is invisible in the separate marginal laws. The resulting finite-dimensional distributions have smooth compact densities, but their supports retain exact algebraic geometry. Generalized Vandermonde minors turn the support into a Schur sum; Littlewood’s identity turns that sum into a pseudohyperbolic product; and a Cauchy determinant turns the same product into a correlation determinant. The equality

$$\text{Vol } Z_Q = \sqrt{\det R_Q}$$

is the organizing identity of the report.

The hyperbolic coordinate $q = \tanh u$ is intrinsic. It linearizes the geometry into pairwise tanh distances, makes the standardized covariance stationary with kernel $\text{sech}(u - v)$, reveals a Szegő feature map, and converts parameter translation into Gaussianization. Bernoulli cumulants give every non-Gaussian correction, Bell partitions organize the approach to Wick’s rule, and the hyperbolic-secant spectral law produces Euler determinants and Meixner–Pollaczek derivative innovations. Confluence returns to the original Fabius parameter curve and gives exact volumes for all derivative jets.

At $q = 1/2$, these constructions become genuinely Fabius-specific: one coordinate has Fabius CDF, the bivariate support is a dyadic geometric-comb lens, its copula boundary contains the inverse Fabius function explicitly, and its parameter jets have volumes $(4/3)^{\binom{d}{2}}$. The Thue–Morse block identity supplies a discrete multiscale filter, while equally spaced hyperbolic nodes produce finite q -superfactorials. These bridges suggest a coherent next program combining convex support geometry, endpoint transseries, q -products, inverse-function asymptotics, and formal proof.

A A self-contained algebraic route to the volume formula

This appendix isolates the exact algebra needed in [section 4](#).

A.1 Zonotope minors

For vectors $g_1, \dots, g_M \in \mathbb{R}^d$, the zonotope

$$Z = \sum_{k=1}^M [0, g_k]$$

has volume

$$\text{Vol}_d Z = \sum_{1 \leq i_1 < \dots < i_d \leq M} |\det[g_{i_1} \cdots g_{i_d}]|. \quad (135)$$

One proof decomposes a generic zonotope into half-open parallelotopes indexed by bases of the associated vector matroid [8]. Degenerate configurations follow by continuity. Formula (135) is often called Shephard's volume formula.

A.2 Generalized Vandermonde to Schur

Let $0 \leq n_1 < \dots < n_d$ and define λ by (21). Reversing the columns of $[x_i^{n_j}]$ and using the bialternant definition of the Schur polynomial gives

$$\det[x_i^{n_j}]_{i,j=1}^d = (-1)^{\binom{d}{2}} \det[x_i^{\lambda_j + d - j}]_{i,j=1}^d \quad (136)$$

$$= (-1)^{\binom{d}{2}} s_\lambda(x) \det[x_i^{d-j}]_{i,j=1}^d \quad (137)$$

$$= s_\lambda(x) \prod_{i < j} (x_j - x_i). \quad (138)$$

For $0 < x_1 < \dots < x_d$, every factor is positive.

A.3 Littlewood's sum

The classical identity [10, 11]

$$\sum_{\ell(\lambda) \leq d} s_\lambda(x_1, \dots, x_d) = \prod_{i=1}^d (1 - x_i)^{-1} \prod_{i < j} (1 - x_i x_j)^{-1} \quad (139)$$

can be proved by the theory of symmetric functions, by a tableau involution, or by antisymmetrizing the geometric series in strictly increasing exponents. For completeness, the last route begins with

$$\prod_{i < j} (x_j - x_i) \sum_{\lambda} s_\lambda(x) = \sum_{0 \leq n_1 < \dots < n_d} \det[x_i^{n_j}]. \quad (140)$$

Expanding the determinant, summing each nested geometric progression, and collecting the common antisymmetric numerator gives

$$\sum_{0 \leq n_1 < \dots < n_d} \det[x_i^{n_j}] = \frac{\prod_{i < j} (x_j - x_i)}{\prod_i (1 - x_i) \prod_{i < j} (1 - x_i x_j)}. \quad (141)$$

Division by the Vandermonde proves (139). Absolute convergence for $|x_i| < 1$ justifies all rearrangements.

Combining (135), (138), and (139) gives (24) in one line after the row factors $1 - q_i$ cancel.

B Mixed moments as colored Bell sums

Suppose a word $w = (i_1, \dots, i_m)$ records coordinate labels. For a block $B \subset [m]$, define its color count

$$\alpha_j(B) = \#\{a \in B : i_a = j\}. \quad (142)$$

Then (15) gives the block weight

$$W(B) = \frac{B_{|B|}^+}{|B|} \frac{\prod_j (1 - q_j)^{\alpha_j(B)}}{1 - \prod_j q_j^{\alpha_j(B)}}. \quad (143)$$

The exact moment is

$$\mathbb{E} \prod_{a=1}^m X_{q_{i_a}} = \sum_{\pi \in \mathcal{P}([m])} \prod_{B \in \pi} W(B). \quad (144)$$

For centered variables, singleton blocks are forbidden and all odd blocks vanish. This representation is particularly efficient when many labels repeat, because partitions can be grouped by colored block type. It is also suited to formalization: every analytic issue has already been discharged in the geometric-series derivation of $W(B)$, leaving a finite combinatorial identity.

For the standardized process, replace the coordinate coefficient $(1 - q_j)q_j^n$ by $\sqrt{1 - q_j^2}q_j^n$ and B_m^+/m by $\kappa_m(\epsilon_0)$. This gives

$$\kappa(Y_{q_{i_1}}, \dots, Y_{q_{i_m}}) = \kappa_m(\epsilon_0) \frac{\prod_{a=1}^m \sqrt{1 - q_{i_a}^2}}{1 - \prod_{a=1}^m q_{i_a}}, \quad (145)$$

which is (64) before the substitution $q_i = \tanh u_i$.

C Fourier regularity in more detail

Choose disjoint blocks

$$B_\ell = \{N_\ell, N_\ell + 1, \dots, N_\ell + d - 1\}, \quad \ell = 1, \dots, M.$$

Let V_ℓ be the $d \times d$ matrix with columns v_n , $n \in B_\ell$. It is invertible. Hence for every $t \in \mathbb{R}^d$,

$$\max_{n \in B_\ell} |t \cdot v_n| \geq \frac{\sigma_{\min}(V_\ell)}{\sqrt{d}} \|t\|. \quad (146)$$

Since $|\operatorname{sinc}_{\text{rad}} x| \leq \min(1, 1/|x|)$, the product of characteristic factors over block B_ℓ is at most

$$\frac{C_\ell}{1 + \|t\|}. \quad (147)$$

All other factors have modulus at most one. Multiplying (147) over M blocks gives

$$|\Phi_Q(t)| \leq C_M (1 + \|t\|)^{-M}. \quad (148)$$

Because M is arbitrary, $t^\beta \Phi_Q(t)$ is integrable for every multi-index β . Fourier inversion then yields a globally C^∞ density, and differentiation under the integral gives

$$\partial^\beta f_Q(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} (-it)^\beta e^{-it \cdot x} \Phi_Q(t) dt. \quad (149)$$

This proof is robust under replacement of uniform digits by any compactly supported digit law whose characteristic function has a first-order decay direction after a full-rank block.

D Artifact and data manifest

The ZIP archive accompanying this report contains:

Path	Description
common_digit_fabius_zonoids.tex	Complete LaTeX source.
common_digit_fabius_zonoids.pdf	Compiled report.

Path	Description
code/experiments.py	Commented reproducibility script.
generated/volume_convergence.csv	Finite-zonotope volume convergence.
generated/gram_volume_identity.csv	Support-volume/correlation-determinant checks.
generated/jet_volume_convergence.csv	Confluent jet-volume checks.
generated/monte_carlo_covariance.csv	Shared-digit Monte Carlo covariance experiment.
generated/euler_hankel_determinants.csv	Exact symbolic Euler determinant checks.
generated/gaussianization_cumulants.csv	Exact cumulants and leading hyperbolic asymptotics.
generated/legendre_coefficients.tex	SymPy-generated exact low-degree coefficient table.
generated/*.pdf, generated/*.png	Report figures in vector and raster formats.
generated/exact_summary.txt	Versions, random seed, and representative identities.
README.md	Build and file instructions.
SOURCE_AUDIT.md	Repository-audit and novelty-boundary notes.
requirements.txt	Minimal Python dependencies.

Acknowledgment of source boundaries

The authoring process used the live documents and thematic manifest in the requested GitHub documentation tree, including subdirectories, to avoid repeating the repository’s established one-parameter theory. The new results are described as repository-novel, not as a blanket claim of first publication. Classical ingredients are credited below.

References

- [1] V. Reshetnikov, *Fabius Function and Rvachev Up-Function*, primary synthesis in the ProveIt repository, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.
- [2] V. Reshetnikov, *Semi-formalized research frontiers for the Fabius/Rvachev program*, canonical compilation and thematic drafts in the ProveIt repository, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers>.
- [3] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Z. Wahrscheinlichkeitstheorie verw. Gebiete* **5** (1966), 173–174.
- [4] B. Jessen and A. Wintner, Distribution functions and the Riemann zeta function, *Trans. Amer. Math. Soc.* **38** (1935), 48–88.
- [5] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, preprint, arXiv:1702.05442, 2017.
- [6] J. Arias de Reyna, Arithmetic of the Fabius function, *Integers* **18** (2018), Paper A51.

- [7] J. K. Haugland, Evaluating the Fabius function, preprint, arXiv:1609.07999, 2016.
- [8] G. C. Shephard, Combinatorial properties of associated zonotopes, *Canad. J. Math.* **26** (1974), no. 2, 302–321.
- [9] E. Gover and N. Krikorian, Determinants and the volumes of parallelotopes and zonotopes, *Linear Algebra Appl.* **433** (2010), no. 1, 28–40.
- [10] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.
- [11] R. P. Stanley, *Enumerative Combinatorics*, vol. 2, 2nd ed., Cambridge University Press, 2012.
- [12] A. Prékopa, On logarithmic concave measures and functions, *Acta Sci. Math. (Szeged)* **34** (1973), 335–343.
- [13] C. de Boor, K. Höllig, and S. Riemenschneider, *Box Splines*, Springer, 1993.
- [14] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.
- [15] R. Koekoek, P. A. Lesky, and R. F. Swarttouw, *Hypergeometric Orthogonal Polynomials and Their q -Analogues*, Springer, 2010.
- [16] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Adv. Comput. Math.* **5** (1996), 329–359.
- [17] P. Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley, 1999.
- [18] J.-P. Kahane, *Some Random Series of Functions*, 2nd ed., Cambridge University Press, 1985.